

Starting the Qualtrics LLM chat: step-by-step

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What you are starting

Three separate programs have to be running at the same time for the chat to work in your survey:

- **Ollama** runs the AI model (Qwen3.5 9B) on your Mac.
- **The chat server** (uvicorn) sits between Qualtrics and Ollama. It holds the AI's instructions, sends the page content, and saves every conversation.
- **cloudflared** gives your Mac a temporary public web address, because the survey runs in your participants' browsers and they cannot reach your laptop directly.

Each one runs in its own Terminal tab and keeps running until you close it. Start them in the order below, because each depends on the one before it. Once you know the routine it takes about two minutes.

Before you start

Know where your folder is. Everything happens in the folder that contains `app.py`. If you are not sure where it is, find it in Finder first and leave that window open. You will drag the folder into Terminal rather than typing its path.

Opening Terminal. Press Cmd+Space, type `terminal`, press Enter. To open a new tab in the same window, press Cmd+T. You will end up with three tabs, one per program.

Moving to your folder. The command `cd` means "change directory." Type `cd` followed by a space, drag the folder from Finder into the Terminal window (the path fills itself in), then press Enter. To confirm you are in the right place, type `ls` and press Enter: you should see `app.py` in the list.

Leave the tabs alone once started. Each program takes over its tab and keeps printing as it runs. That is normal. Closing a tab or pressing Ctrl+C stops that program, and the chat stops working for participants.

Keep the Mac awake. If your Mac sleeps, all three stop. Either plug it in and set the display to never sleep in System Settings, or run `caffeinate -dims` in a fourth tab and leave it open.

Step 1: Start Ollama

This is the AI model itself. Use the first Terminal tab.

```
ollama serve
```

Press Enter. You will see lines of startup text, and then it sits there. Leave this tab open.

What success looks like: the tab stops scrolling and does not return you to a normal prompt. To double-check, open <http://127.0.0.1:11434> in your browser; it should say that Ollama is running.

If it says `address already in use`: Ollama is already running, probably from the menu-bar app. Nothing to fix. Move on to Step 2.

If it says `command not found`: open the Ollama app once from your Applications folder, which installs the command-line tool, then try again.

You do not need to load the model yourself. It loads automatically when the first participant message arrives, which makes that first reply slower than the rest.

Step 2: Start the chat server

Open a new tab with Cmd+T.

a. Go to your folder. Type `cd` and a space, drag the folder holding `app.py` into the window, press Enter.

b. Turn on the virtual environment. This is the private Python setup where the server's packages live.

```
source .venv/bin/activate
```

If that says "No such file or directory," the environment is one folder up, so use `source ../.venv/bin/activate` instead.

What success looks like: `(.venv)` appears at the start of the prompt line.

c. Start the server.

```
uvicorn app:app --host 127.0.0.1 --port 8000
```

What success looks like: several startup lines ending with something like `Uvicorn running on http://127.0.0.1:8000`. The tab then sits quietly, and later prints a line each time a participant sends a message. Leave it open.

If you see a red traceback instead, the most common causes are being in the wrong folder (no `app.py` there) or forgetting the `source` command, so the packages are missing.

Step 3: Start the public address

Open a third tab with `Cmd+T`. You do not need the virtual environment or any particular folder for this one.

```
cloudflared tunnel --url http://127.0.0.1:8000
```

After a few seconds it prints a box containing a web address ending in `.trycloudflare.com`. That address is how participants' browsers reach your Mac.

Copy it now. Select the address with your mouse and press `Cmd+C`. If the box has scrolled out of view, scroll up in the tab to find it.

Leave this tab open too. If it closes, the address stops working immediately.

Important: this address is new every single time you start `cloudflared`. Whenever you restart, you must paste the new one into Qualtrics (Step 4). A named Cloudflare tunnel gives you a permanent address and removes this step; worth setting up before real data collection.

Step 4: Put the address into Qualtrics

1. Open your survey in Qualtrics and click the chat question.
2. In the left panel, scroll to **JavaScript** (or use the gear icon on the question, then **Add JavaScript**).
3. Near the top of the code, find the line that begins `var SERVER_URL =`.
4. Replace the address between the quotation marks with the one you just copied.

It should end up looking like this, with no slash at the end:

```
var SERVER_URL = "https://random-words-here.trycloudflare.com";
```

5. Click **Save**, then publish the survey if participants are taking it live. Preview mode uses your unpublished edits, but the live link does not.

While you are in there, check the other settings if anything has changed: `API_KEY` must match what the server uses, and `MIN_TURNS` and `MAX_TURNS` control how many messages a participant sends.

Step 5: Test before anyone arrives

a. Check the server. In your browser, open your address with `/health` on the end:

```
https://random-words-here.trycloudflare.com/health
```

You want to see `{"ok":true,"model":"qwen3.5:9b","model_available":true}`.

- `ok: false` means the chat server is running but cannot reach Ollama. Check the tab from Step 1.
- `model_available: false` means Ollama is running but the model is not downloaded. Run `ollama pull qwen3.5:9b`.
- Visiting the address **without** `/health` gives `{"detail":"Not Found"}`. That is normal, not an error; there is simply no page at the root.

b. Chat in Preview. Click **Preview** in Qualtrics, go to the chat question, and send a message. The first reply takes longer while the model loads. Watch the uvicorn tab: a new line should appear for each message.

c. Check the data saved. Finish the preview survey to the end, then go to **Data & Analysis** and look at the `chat_log` (or `__js_chat_log`) column for your preview response.

When the session is over

Stopping. In each tab, press Ctrl+C (hold Control, press C). Then close the tabs. Order does not matter. Once cloudflared stops, the chat stops working for anyone still in the survey, so make sure nobody is mid-session.

Leaving the virtual environment. In the uvicorn tab, type `deactivate`. The `(.venv)` disappears. This is optional; closing the tab does the same thing.

Back up your data. The full record of every conversation lives in `chat_logs.sqlite3`, in the same folder as `app.py`. It exists only on your Mac. After each session, copy it somewhere safe, such as a university drive, following whatever your IRB protocol says

about where this data may live.

Export to spreadsheets when you want to look at it. In a tab in that folder, with `(.venv)` active:

```
python3 export_logs.py
```

That writes `messages.csv` (one row per message) and `sessions.csv` (one row per conversation). Merge them with your Qualtrics export by matching `response_id` to Qualtrics's `ResponseId`.

When something goes wrong

What you see	What it means	What to do
<code>{"detail": "Not Found"}</code> in the browser	Everything is working; there is just no page at the root address	Add <code>/health</code> to the end of the address
<code>command not found</code>	That program is not installed, or its tab is missing the setup	Ollama: open the app once. cloudflared: <code>brew install cloudflared</code> uvicorn: run the <code>source .venv/bin/activate</code> line first
<code>address already in use</code>	Something is already running on that port	For Ollama, carry on. For uvicorn, an older server is still running; find its tab and press Ctrl+C
<code>"ok": false</code> at <code>/health</code>	The chat server cannot reach Ollama	Check the Ollama tab is still running
<code>"model_available": false</code>	The model is not downloaded	<code>ollama pull qwen3.5:9b</code>
Error 502 or 1033 in the browser	cloudflared is running but uvicorn is not	Check the uvicorn tab for a crash and restart it
The chat box appears but replies never come	Usually the address in Qualtrics no longer matches the current tunnel	Recheck <code>SERVER_URL</code> against the address in the cloudflared tab

What you see	What it means	What to do
No chat box at all in the survey	The JavaScript did not save, or you are on a different question	Reopen the question's JavaScript and confirm the code is there
Replies take 20+ seconds	The model is loading, or your Mac is working hard	Normal on the first message; if it persists, lower <code>LLM_MAX_TOKENS</code>

If a Terminal tab shows a long red error, the last few lines are the useful part. Copy those rather than the whole block when asking for help.

Quick reference

Once the routine is familiar, this is the whole thing. Three tabs, then two checks.

Tab 1

```
ollama serve
```

Tab 2 (Cmd+T)

```
cd /path/to/your/folder
source .venv/bin/activate
uvicorn app:app --host 127.0.0.1 --port 8000
```

Tab 3 (Cmd+T)

```
cloudflared tunnel --url http://127.0.0.1:8000
```

Then: copy the `.trycloudflare.com` address, paste it into `SERVER_URL` in the Qualtrics JavaScript, save, and open the address with `/health` on the end to confirm `"ok":true`.

To stop: Ctrl+C in each tab. To back up: copy `chat_logs.sqlite3` somewhere safe.